6COM2005 Practical Coursework 50%
Data Mining
Semester AB 2025/2026

Due date: January 21, 2026 at 11:00 AM

This coursework is worth 50% of the overall grade. There are 100 marks to achieve, each translating to 0.5% of your overall module grade.
There are five tasks in total. The first three are the main tasks, and for each of these, you will have two options to choose from. Each option carries the same number of marks and aligns with the later tasks, so choose the one you feel most confident with. The fourth task is the comparison and discussion of your obtained results. The final task is a demonstration and viva, where you will present and defend your work.

Submission Guidelines

You may discuss your general ideas and thoughts with peers, but the work submitted must be distinctly yours and your own.

1. Accept the Git assignment/coursework: Git Assignment
2. Clone the Git repository, which contains datasets and a template notebook to begin your work. You will complete this coursework using the provided coursework notebook (.ipynb) file.
3. Git activity is important: (i), your repository will be monitored and will contribute to your overall grade. (ii), your Git history will show how many days you worked on the coursework and the version history of how you improved your work.
4. After completing all parts, download your final Jupyter Notebook (.ipynb) and submit it via Canvas. ccording to UH policy, only submissions via Canvas will be considered.

Use of Generative AI

In this assessment you are permitted to use genAI tools (or a proofreader or proofreading service) to proofread your work but not permitted to use AI tools in the creation of your work. This category will apply irrespective of the fact that the grading criteria include credit for English and grammar.

General UH guidance on generative AI advice to students can be found using Ask Herts.

Additional information

1. Please familiarise yourself with regulations governing assessment offences including Plagiarism and Collusion (UPR AS14).
2. Engage with guidance for avoiding plagiarism via the University SkillUp module and 1-2-1 sessions with our specialist SASH team.
3. A score of 40% or above represents a pass performance at honours level.
4. Late submission of any item of coursework for each day or part thereof (or for hard copy submission only, working day or part thereof) for up to five days after the published deadline, coursework relating to modules at Levels 0, 4, 5, 6 submitted late (including deferred coursework, but with the exception of referred coursework), will have the numeric grade reduced by 10 grade points until or unless the numeric grade reaches or is 40. Where the numeric grade awarded for the assessment is less than 40, no lateness penalty will be applied.

Learning Outcomes

This coursework assesses the following module learning outcomes

1. Demonstrate knowledge and a critical understanding of the key functionality of various data mining components and the functionality, strengths, and applicability of various data mining models, approaches, and algorithms.
2. Demonstrate knowledge and a critical understanding of the techniques for designing data mining systems.
3. Apply a range of data mining methods and algorithms to a variety of problems.
4. Critically evaluate different algorithms and methods in data mining, interpret the output from data mining tasks, and present clear outcomes and indications from data sets by suitable visualization.

Task 1: Prepare the data set [15 marks]

1. Choose a dataset of your choice. Load the data set as a dataframe [1 marks].
2. Clean the dataset and convert text (categorical) features to numbers [2 marks]
3. Use both numerosity reduction as well as feature reduction so that your data set only has 3 features (The class column does not count). [6 marks]
4. Create a markdown cell and justify how you used both numerosity reduction and feature reduction to reduce your dataset to only three features (excluding the class column), and explain why it is important for your analysis. [3 marks, word count max 150 words]

   Note: Commit and push changes to the Git repository after completing each sub-task with a clear and descriptive commit message. If you revisit Part-A and make any improvements or modifications to the code, clearly explain them in the commit message before pushing the changes [3 marks].

Task 2: Clustering [25 marks]

Choose one of the following Clustering algorithms: k-means or DBSCAN. Make sure you remove the class column from the data set before clustering.

K-Means

1. Implement the K-means algorithm to work with the cleaned data set. You can use parts of your implementation you created during the practical but will need to adapt it to work with 3-dimensional data [6 marks].
2. Choose the right number of centroids for your dataset and justify your choice in the markdown [3 marks].
3. Run the algorithm 3 times and store the results, so that it is clear which point belongs to which centroid [4 marks].
4. use appropriate plots to visualize the obtained results [3 marks].
5. Now use a library to implement the K-means algorithm on your 3-dimensional data. You should get the same results [3 marks].
6. In a markdown cell, create a table for each run that contains the final position of each centroid and shows the count of data points assigned to each cluster after the run. Interpret and discuss your obtained results.[3 marks, word count max 150 words]

Note: Commit and push changes to the Git repository after completing each sub-task with a clear and descriptive commit message. If you revisit to make any improvements or modifications to the code, clearly explain them in the commit message before pushing the changes [3 marks].

DBSCAN

1. Implement the DBSCAN algorithm to work with the cleaned data set. You can use parts of your implementation you created during the practical but will need to adapt it to work with 3-dimensional data [6 marks].

2. Choose 3 sets of parameters (ϵ and MinPoints) to run the algorithm with. State these in your report and justify why you chose these specific values [3 marks].

3. Run the algorithm 3 times with the different parameter sets and store the results in one or multiple csv-files [4 marks].

4. use appropriate plots to visualize the obtained results [3 marks].

5. Now use a library to implement the K-means algorithm on your 3-dimensional data. You should get the same results [3 marks].

6. In a mark down cell, create a table for each run that contains the count of core, border and noise points after each run. In a different table, list the count of data points assigned to the different clusters after the run. You may have differently many clusters for each run? Interpret and discuss your obtained results. [3 marks, word count max 150 words]

   Note: Commit and push changes to the Git repository after completing each sub-task with a clear and descriptive commit message. If you revisit to make any improvements or modifications to the code, clearly explain them in the commit message before pushing the changes [3 marks].

Task 3: Classification [25 marks]

Choose one of the following Classification Algorithms: K-nearest neighbour or Gaussian naive Bayes.

K-Nearest Neighbour

1. Implement the K-Nearest Neighbor algorithm to work with the cleaned data set. You can use parts of your implementation you created during the practical but will need to adapt it to work with 3-dimensional data [5 marks].

2. Obtain the correlation matrix and plot each feature against the the target feature. Discuss the relationship in a mark down cell. use subplot [2 marks]

3. Choose 3 different values for k to create the model with. Split your data into training and test data. State these in a mark down cell and justify why you chose these specific values [4 marks].

4. Create the 3 models with the different k using your training data [3 marks].

5. Use the test data to evaluate your resulting classifier using the confusion matrix and accuracy [3 marks].

6. Now use a library to implement the K-Nearest Neighbour on your 3-dimensional data. You should get the same results [2 marks].

7. Interpret and discuss your obtained results. [3 marks, word count max 150 words]

   Note: Commit and push changes to the Git repository after completing each sub-task with a clear and descriptive commit message. If you revisit to make any improvements or modifications to the code, clearly explain them in the commit message before pushing the changes [3 marks].

Gaussian Naive Bayes

1. Implement the Gaussian Naive Bayes algorithm to work with the cleaned data set. You can use parts of your implementation you created during the practical but will need to adapt it to work with 3-dimensional data [5 marks].

2. Choose 2 different ways to split your data into training and test data. State these in your report and justify why you chose these specific values [4 marks].

3. Create the 2 Gaussian Naive Bayes models with the different training data sets [4 marks].

4. Use the different test data to evaluate your resulting classifiers using the confusion matrix and accuracy. Make sure you use the correct test set for the classifier [4 marks].

5. Now use a library to implement the Gaussian Naive Bayes algorithm on your 3-dimensional on both training dataset. You should get the same results [2 marks].

6. Interpret and discuss your obtained results. [3 marks, word count max 150 words]

   Note: Commit and push changes to the Git repository after completing each sub-task with a clear and descriptive commit message. If you revisit to make any improvements or modifications to the code, clearly explain them in the commit message before pushing the changes [3 marks].

Task 4: Comparison and Discussion [15 marks]

Lastly, in a mark down call, compare your clustering results with the classification results.
This is a very general task, and there are many things you can notice and discuss. For example: you could discuss the choice of parameters, what happens when the number of clusters matches the number of classes (or not), specific data points that are difficult to cluster or classify and why and many more.

I expect you to discuss three main points in total, using about 200–300 words. If you can include four points without making them too brief, that is fine. If you only cover two points but explain them in depth, that is also acceptable. Just make sure your answer stays within the 200–300 word limit.

Marking scheme: If you discuss four points, you can earn up to 4 marks per point. If you discuss three points, you can earn up to 5 marks per point. If you discuss two points in depth, you can earn up to 7 marks per point.

Commit and push changes to the Git repository after completing [1 mark]

Create a markdown cell and clearly state the name of the AI tool you used and explain its purpose. Describe how you reviewed, adapted, or integrated the AI-generated output into your own work [maximum 100 words]. This section carries no marks but is mandatory to include.

Task 5: Demonstration and Viva [20 marks]

Make sure you download and submit the final notebook on Canvas. According to the UH policy, only submissions made through Canvas will be considered. After the due date, you will appear for a demonstration within the following two weeks.

1. You will have up to 10 minutes to demonstrate the Jupyter Notebook you have created [10 marks].
   Marking scheme:: Marks will be awarded for the correct demonstration of concepts, justification of your choices, critical evaluation of your results, professionalism, and confidence.

2. After the demonstration, you will have a 5-minute viva where you will be asked questions about your work [10 marks].
   Marking scheme: Marks will be awarded for the correct understanding of concepts, clear justifications, and accurate interpretation of your results.

Submission Checklist

1. Commit and push your work to the Git repository.
2. Download and submit the final notebook (.ipynb and .html) on Canvas. According to UH policy, only submissions made through Canvas will be considered.
3. Arrange a time with the module leader and attend the demonstration.

Any questions, please ask.

All the best!